

IX. INTRODUCTION TO HONEYPOTS

Aim:

A guided room covering the deployment of honeypots and analysis of botnet activities

Procedure:

1. Understand the concept and purpose of honeypots in network security.
2. Set up a basic honeypot to simulate vulnerable services or systems.
3. Configure the honeypot to log and monitor incoming traffic and interactions.
4. Deploy the honeypot within a controlled network environment.
5. Observe and analyze attacker behavior and attempted exploits.
6. Record and evaluate the collected data for security insights.

Tasks:

Task 1: Introduction

Deploy the demo machine

No answer needed

✓ Correct Answer

Task 2: Types of Honeypots

Read and understand the above

No answer needed

✓ Correct Answer

Task 3: Cowrie Demo

Try running some commands in the honeypot

No answer needed

✓ Correct Answer

Create a file and then log back in is the file still there? (Yay/Nay)

Nay

✓ Correct Answer

Task 4: Cowrie Logs

Have a look through the logs and see how the activity from the last task has been recorded by the system.

No answer needed

✓ Correct Answer

Task 5: Attacks against SSH

How many passwords include the word "password" or some other variation of it e.g "p@ssw0rd"

15

✓ Correct Answer



What is arguably the most common tool for brute-forcing SSH?

hydra

✓ Correct Answer

What intrusion prevention software framework is commonly used to mitigate SSH brute-force attacks?

Fail2Ban

✓ Correct Answer

Task 6: Typical Bot Activity

What's the full model name of CPU does the honeypot "use"?

Intel(R) Core(TM) i9-11900KB CPU @ 3.30GHz

✓ Correct Answer



Does the honeypot return the correct values when `uname -a` is run? (Yay/Nay)

Nay

✓ Correct Answer



What flag must be set to pipe `wget` output into bash?

-O

✓ Correct Answer

How would you disable bash history using `unset`?

unset HISTFILE

✓ Correct Answer

Task 7: Identification Techniques

What brand of device is the bot in the first sample searching for? (BotCommands/Sample1.txt)

Mikrotik

✓ Correct Answer

What are the commands in the second sample changing? (BotCommands/Sample2.txt)

root password

✓ Correct Answer

What is the name of the group that runs the botnet in the third sample? (BotCommands/Sample3.txt)

Outlaw

✓ Correct Answer

Task 8: SSH Tunneling

What application is being targetted in the first sample? (Tunnelling/Sample1.txt)

WordPress

✓ Correct Answer

Is the URL in the second sample malicious? (Tunnelling/Sample2.txt) (Yay/Nay)

Nay

✓ Correct Answer

Task 9: Recap and Extra Resources

Read and understand the above

No answer needed

✓ Correct Answer

Result:

Successfully implemented and analyzed a honeypot by simulating vulnerable systems, monitoring malicious activities, and gaining insights into potential attack patterns and threats.